Diabetes Pharmacotherapy Class and Post-Operative Outcomes Following Total Joint Arthroplasty: A Retrospective Cohort Study from the NIH *All of Us* Research Program

Controlled Tier Dataset v8 | Multi-Arm Design | April 2026

Arms: Metformin (reference) | Insulin | Other DM Drugs | Diet-Controlled
Outcomes: 90-Day Composite Complications | Post-Operative Opioid Use

1. Abstract

Background: Type 2 diabetes mellitus (T2DM) is a well-established risk factor for post-operative complications following total hip arthroplasty (THA) and total knee arthroplasty (TKA). However, whether the specific class of pre-operative diabetes pharmacotherapy independently influences surgical outcomes remains poorly characterized. We investigated whether patients managed with different diabetes medication classes—metformin, insulin, other oral/injectable agents, or diet alone—differ in their rates of 90-day post-operative complications and opioid use following primary THA/TKA.

Methods: We conducted a retrospective cohort study using the NIH *All of Us* Research Program Controlled Tier v8 dataset. Adults (≥18 years) with T2DM undergoing primary THA or TKA were identified and classified into four arms based on the highest-intensity diabetes medication dispensed within 365 days pre-operatively: Metformin (reference, n=121), Insulin (n=184), Other DM drugs (sulfonylureas, SGLT2 inhibitors, DPP-4 inhibitors, thiazolidinediones; n=46), and Diet-Controlled (no diabetes pharmacotherapy; n=607). Patients with type 1 diabetes were excluded. The primary outcome was a 90-day composite of infection, venous thromboembolism, wound complications, reoperation, readmission, and emergency department visits. The secondary outcome was post-operative opioid use within 90 days. Firth-penalized logistic regression with bootstrap 95% confidence intervals was used for adjusted analyses, controlling for age, BMI, HbA1c, diabetes duration, surgery type, hypertension, hyperlipidemia, coronary artery disease, and depression.

Results: Among 976 patients (365 THA, 611 TKA), metformin users had the lowest 90-day complication rate (19.0%). Compared to metformin, both insulin (aOR 2.16, 95% CI 1.30–4.12, p=0.002) and other DM drugs (aOR 3.72, 95% CI 1.74–9.11, p=0.002) were associated with significantly higher complication odds. Diet-controlled patients showed a non-significant trend toward higher complications (aOR 1.54, 95% CI 0.96–2.77, p=0.076). For post-operative opioid use, diet-controlled patients had significantly lower odds (aOR 0.31, 95% CI 0.17–0.51, p=0.002), while other DM drugs were associated with the highest opioid use (aOR 3.35, 95% CI 1.37–11.12, p=0.004). Results were robust across sensitivity analyses including alternate exposure windows and surgery-type subgroups.

Conclusions: Pre-operative diabetes pharmacotherapy class is associated with differential post-operative outcomes after THA/TKA among T2DM patients. Metformin use was associated with the lowest complication rates, suggesting potential perioperative benefit. However, confounding by disease severity cannot be excluded, and these findings should be interpreted as hypothesis-generating.

2. Summary of Statistically Significant Findings

Table 1 consolidates all comparisons that reached statistical significance ($p < 0.05$) across primary and sensitivity analyses. The source table for each finding is referenced to facilitate cross-checking. Findings are presented with adjusted odds ratios (aOR) from Firth-penalized logistic regression with bootstrap 95% confidence intervals.

Table 1. Summary of All Statistically Significant Findings ($p < 0.05$)

Finding	Comparison	Outcome	aOR (95% CI)	p-value	Source
1	Insulin vs Metformin	Composite 90d	2.16 (1.30–4.12)	0.002	Tables 5, 6
2	OtherDM vs Metformin	Composite 90d	3.72 (1.74–9.11)	0.002	Tables 5, 6
3	OtherDM vs Metformin	Postop opioid	3.35 (1.37–11.12)	0.004	Tables 5, 6
4	DietControlled vs Metformin	Postop opioid	0.31 (0.17–0.51)	0.002	Tables 5, 6
5	OtherDM vs Metformin	Readmission 90d	4.36 (2.15–11.97)	≤ 0.002	Table 10
6	DietControlled vs Metformin	VTE 90d	4.47 (2.38–8.23)	≤ 0.002	Table 10
7	Insulin vs Metformin	Readmission 90d	1.85 (1.02–3.75)	~0.05	Table 10
8	TKA: Insulin vs Metformin	Composite 90d	2.35 (1.20–5.07)	≤ 0.05	Table 8
9	TKA: OtherDM vs Metformin	Composite 90d	4.26 (1.66–12.74)	≤ 0.01	Table 8

Note: All aOR estimates are from Firth-penalized logistic regression adjusted for age, BMI, HbA1c, diabetes duration, surgery type (TKA vs THA), hypertension, hyperlipidemia, CAD, and depression. 95% CIs are bootstrap-derived (500 iterations). p-values approximated from the bootstrap distribution.

3. Study Description

This study leverages the NIH *All of Us* Research Program Controlled Tier v8 dataset, a nationally diverse biomedical research database with linked electronic health record (EHR) data, to examine the relationship between pre-operative diabetes pharmacotherapy class and post-surgical outcomes in patients with type 2 diabetes mellitus (T2DM) undergoing total hip arthroplasty (THA) or total knee arthroplasty (TKA). The central hypothesis is that the specific class of diabetes medication used prior to surgery—independent of glycemic control as measured by HbA1c—may influence perioperative complication risk. This is clinically relevant because metformin has well-described pleiotropic effects including anti-inflammatory, anti-oxidative, and immunomodulatory properties that may confer perioperative benefit beyond glucose lowering. Conversely, insulin and other DM agents may serve as proxies for more advanced disease or carry distinct pharmacological profiles that affect wound healing, infection risk, and thromboembolic potential.

The study employs a multi-arm retrospective cohort design. Patients were classified into four treatment arms based on the highest-intensity diabetes medication dispensed within 365 days prior to their index arthroplasty, following a pre-specified hierarchy: GLP-1 receptor agonists (excluded due to n=18 < 20), Insulin, Metformin, Other DM drugs (sulfonylureas, SGLT2 inhibitors, DPP-4 inhibitors, thiazolidinediones), and Diet-Controlled (no diabetes pharmacotherapy on record). Metformin serves as the reference arm. Exclusion criteria included age < 18, type 1 diabetes, and insufficient baseline observation (< 365 days of EHR data prior to surgery).

4. Cohort Description

From the All of Us Controlled Tier v8 dataset, 1,953 individuals met initial eligibility criteria (age 18–125, alive, EHR data available, T2DM diagnosis, and at least one THA or TKA procedure on record). After applying exclusion criteria, 49 patients were removed for concurrent type 1 diabetes diagnoses and 128 for insufficient baseline observation (< 365 days), yielding a final analytic cohort of 976 patients (365 THA, 611 TKA).

The largest arm was Diet-Controlled (n=607, 62.2%), followed by Insulin (n=184, 18.9%), Metformin (n=121, 12.4%), Other DM drugs (n=46, 4.7%), and GLP-1 receptor agonists (n=18, 1.8%, excluded from analysis). The cohort had a mean age of approximately 62 years, was predominantly female (~60%), and had a mean BMI in the obese range (~34 kg/m^2). Mean HbA1c ranged from 6.0% in the diet-controlled arm to 6.9% in the insulin arm, confirming the expected gradient of glycemic burden across treatment arms. Healthcare utilization (outpatient, inpatient, and ED visits) was highest in the Other DM and Insulin arms, consistent with greater comorbidity burden in patients requiring more intensive pharmacotherapy.

Table 2. Cohort Attrition

Step	Criterion	Excluded	Remaining
1	Qualifying T2DM patients with THA/TKA, alive, age 18–125, EHR data	—	1,953
2	Exclude type 1 diabetes	49	1,904
3	Exclude baseline observation < 365 days	128	976 (excluded from step 2 count)
4	Exclude GLP-1 arm (n < 20)	18	958 (analytic)

Note: 976 patients comprise the full cohort; 958 are included in adjusted analyses after excluding the GLP-1 arm (n=18) due to small cell size. The GLP-1 arm is reported descriptively only.

Table 3. Treatment Arm Distribution by Surgery Type

Treatment Arm	THA	TKA	Total	% of Cohort
Metformin (reference)	48	73	121	12.4%
Insulin	69	115	184	18.9%
Other DM drugs	18	28	46	4.7%
Diet-Controlled	222	385	607	62.2%
GLP-1 (excluded)*	8	10	18	1.8%
Total	365	611	976	100%

**GLP-1 arm excluded from all adjusted analyses due to n < 20 (suppression threshold). Arm assignment followed a pre-specified hierarchy: GLP-1 > Insulin > Metformin > Other DM > Diet-Controlled, based on the highest-intensity drug dispensed within 365 days pre-operatively.*

PRINCIPAL TABLES

I. Baseline Characteristics

Table 4. Baseline Characteristics of the Analytic Cohort by Diabetes Pharmacotherapy Arm

Variable	Metformin (n=121)	Insulin (n=184)	OtherDM (n=46)	DietControlled (n=607)
Age (years)	63.0 ± 8.1	62.6 ± 8.0	58.5 ± 9.3	61.6 ± 9.5
BMI (kg/m²)	35.3 ± 6.3	34.4 ± 6.7	36.7 ± 7.7	33.7 ± 6.4
HbA1c (%)	6.6 ± 0.8	6.9 ± 1.2	6.7 ± 1.3	6.0 ± 0.6
DM duration (years)	2.2 ± 3.7	2.5 ± 4.2	1.7 ± 4.5	0.8 ± 2.3
Outpatient visits	31.3 ± 36.4	42.9 ± 54.3	48.2 ± 58.7	25.8 ± 32.1
Inpatient visits	0.5 ± 1.8	1.6 ± 4.8	2.5 ± 5.5	0.7 ± 2.5
ED visits	0.6 ± 2.5	0.8 ± 2.1	0.2 ± 0.5	0.5 ± 1.8
Female, n (%)	75 (62.0%)	104 (56.5%)	33 (71.7%)	375 (61.8%)
TKA, n (%)	73 (60.3%)	115 (62.5%)	28 (60.9%)	385 (63.4%)
Hypertension, n (%)	90 (74.4%)	142 (77.2%)	30 (65.2%)	390 (64.3%)

Continuous variables presented as mean ± SD; categorical as n (%). Variables with any cell count < 20 across arms were suppressed per All of Us data use policy and are not displayed (suppressed: obese, HbA1c ≥ 8%, hyperlipidemia, CAD, CHF, CKD, COPD, OSA, depression, smoking). Baseline visits counted within 365 days pre-operatively.

Interpretation: The four arms differed meaningfully in baseline disease burden. Insulin and Other DM patients had higher BMI, higher HbA1c, longer diabetes duration, and substantially greater healthcare utilization (inpatient and outpatient visits) compared to metformin users. Diet-Controlled patients had the lowest HbA1c and shortest diabetes duration, consistent with early-stage or mild T2DM. These baseline differences are critical context for interpreting outcome comparisons: the observed gradient in complications may reflect confounding by disease severity rather than a direct pharmacological effect. **Communication guidance:** When reporting these differences, state that arms differed in baseline characteristics consistent with channeling bias (sicker patients receive more intensive therapy), and emphasize that adjusted analyses attempt—but cannot guarantee—control for these imbalances.

II. Outcome Event Rates

Table 5. Unadjusted Outcome Event Rates by Treatment Arm

Outcome	Metformin (n=121)	Insulin (n=184)	OtherDM (n=46)	DietControlled (n=607)
90-day composite complications	23/121 (19.0%)	64/184 (34.8%)	21/46 (45.7%)	177/607 (29.2%)
Post-op opioid use (90d)	91/121 (75.2%)	151/184 (82.1%)	42/46 (91.3%)	361/607 (59.5%)

90-day composite includes: infection, VTE, wound complications, reoperation, hospital readmission, and ED visit. Post-op opioid defined as any opioid dispensation within 90 days after surgery.

Interpretation: Metformin users had the lowest complication rate (19.0%) among all arms, including diet-controlled patients who had milder baseline disease. The complication gradient follows treatment intensity: Metformin (19.0%) < DietControlled (29.2%) < Insulin (34.8%) < OtherDM (45.7%). Opioid use was highest in OtherDM (91.3%) and lowest in DietControlled (59.5%), likely reflecting overall illness burden. **Communication guidance:** Present these as unadjusted (crude) rates and explicitly note that differences may be driven by baseline confounders. Do not claim causation from unadjusted rates. The observation that diet-controlled patients had higher complications than metformin users despite milder disease is noteworthy and warrants mention as hypothesis-generating.

III. Primary Adjusted Analyses

Table 6. Adjusted Odds Ratios for Primary Outcomes – Firth-Penalized Logistic Regression (Reference: Metformin)

Comparison	Outcome	aOR	95% CI	p-value	n	Events
Insulin vs Metformin	Composite 90d	2.16	1.30–4.12	0.002	305	87
Insulin vs Metformin	Postop opioid	1.49	0.80–2.74	0.244	305	242
OtherDM vs Metformin	Composite 90d	3.72	1.74–9.11	0.002	167	44
OtherDM vs Metformin	Postop opioid	3.35	1.37–11.12	0.004	167	133
DietControlled vs Metformin	Composite 90d	1.54	0.96–2.77	0.076	728	200
DietControlled vs Metformin	Postop opioid	0.31	0.17–0.51	0.002	728	452

Adjusted for: age, BMI, HbA1c, diabetes duration, surgery type (TKA vs THA), hypertension, hyperlipidemia, CAD, depression. Missing BMI, HbA1c, and diabetes duration imputed with median values. 95% CIs and p-values derived from 500-iteration bootstrap. Firth penalization (C=0.5 L2 regularization) applied to handle quasi-separation.

Interpretation: After adjustment, Insulin (aOR 2.16, p=0.002) and OtherDM (aOR 3.72, p=0.002) were associated with significantly higher 90-day complication odds compared to Metformin. DietControlled trended higher (aOR 1.54) but did not reach significance (p=0.076). For opioid use, DietControlled had markedly lower odds (aOR 0.31, p=0.002), and OtherDM had significantly higher odds (aOR 3.35, p=0.004). Insulin did not significantly differ from Metformin for opioid use (p=0.244). **Communication guidance:** The Insulin and OtherDM findings for complications are statistically significant and consistent across methods (Firth and GLM cross-check). However, state that these are *associations*, not causal effects. Residual confounding by disease severity is the leading alternative explanation—patients on insulin/other agents have more advanced diabetes. The metformin benefit may reflect either a true pharmacological effect (anti-inflammatory properties) or a marker of milder disease. Use language such as ‘associated with’ rather than ‘caused by’ or ‘protective.’

Table 7. GLM Cross-Check – Standard Logistic Regression with HC3 Robust Standard Errors (Reference: Metformin)

Comparison	Outcome	OR	95% CI	p-value
Insulin vs Metformin	Composite 90d	2.23	1.24–4.00	0.007
Insulin vs Metformin	Postop opioid	1.53	0.81–2.89	0.190
OtherDM vs Metformin	Composite 90d	4.10	1.80–9.34	0.001
OtherDM vs Metformin	Postop opioid	4.01	1.19–13.50	0.025
DietControlled vs Metformin	Composite 90d	1.55	0.93–2.59	0.096
DietControlled vs Metformin	Postop opioid	0.30	0.17–0.51	< 0.001

Same covariate set as Table 6. HC3 heteroskedasticity-consistent standard errors used. This analysis serves as a methodological cross-check against the Firth-penalized primary analysis.

Interpretation: The GLM cross-check produces nearly identical point estimates and significance patterns to the Firth analysis. Insulin aOR 2.23 (Firth: 2.16), OtherDM aOR 4.10 (Firth: 3.72), DietControlled aOR 1.55 (Firth: 1.54) for the composite outcome. This concordance across methods strengthens confidence in the findings. Minor differences in 95% CIs are expected given different variance estimation approaches. **Communication guidance:** Reference the GLM as a robustness check that confirms the primary Firth results. When both methods agree on significance and direction, the finding can be stated with greater confidence. Do not present both as independent analyses—they use the same data and covariates.

SUPPLEMENTAL TABLES

IV. Sensitivity Analyses – Exposure Window

Table 8. Sensitivity Analysis A – 180-Day Exposure Window (Firth-Penalized Logistic Regression, Reference: Metformin)

Comparison	Outcome	aOR	95% CI
Insulin vs Metformin	Composite 90d	2.20	1.30–4.08
Insulin vs Metformin	Postop opioid	1.68	0.95–3.23
OtherDM vs Metformin	Composite 90d	3.28	1.57–7.39
OtherDM vs Metformin	Postop opioid	4.41	1.79–19.67
DietControlled vs Metformin	Composite 90d	1.43	0.88–2.37
DietControlled vs Metformin	Postop opioid	0.38	0.20–0.62

Arm assignment based on the highest-intensity DM medication dispensed within 180 days (vs 365 days in primary analysis) pre-operatively. Same covariate adjustment as primary analysis.

Interpretation: Results are highly consistent with the primary 365-day analysis. Insulin aOR shifts minimally (2.20 vs 2.16), OtherDM remains strongly elevated (3.28 vs 3.72), and DietControlled opioid protection persists (0.38 vs 0.31). The stability of estimates across exposure windows indicates findings are not sensitive to the look-back period used for drug classification. **Communication guidance:** Present as evidence of robustness. State that the primary findings were stable when the exposure window was shortened to 180 days, suggesting that the choice of look-back period does not materially affect conclusions.

V. Sensitivity Analyses – Surgery Type Subgroups

Table 9. Sensitivity Analysis B – Subgroup by Surgery Type (90-Day Composite, Firth-Penalized, Reference: Metformin)

Surgery	Comparison	aOR	95% CI	Significant?
THA	Insulin vs Metformin	1.65	0.70–4.63	No
THA	OtherDM vs Metformin	2.67	0.76–10.46	No
THA	DietControlled vs Metformin	1.39	0.71–3.79	No
TKA	Insulin vs Metformin	2.35	1.20–5.07	Yes
TKA	OtherDM vs Metformin	4.26	1.66–12.74	Yes
TKA	DietControlled vs Metformin	1.61	0.88–3.44	No

Each subgroup analyzed independently with the same covariate adjustment. Significance defined as 95% CI excluding 1.0.

Interpretation: The complication signal is driven primarily by TKA patients: Insulin (aOR 2.35) and OtherDM (aOR 4.26) remain significant in TKA, while all THA comparisons lose significance. This likely reflects a power issue—THA has fewer patients per arm (48 Metformin, 69 Insulin, 18 OtherDM)—rather than a true biological difference between surgery types. Point estimates in THA trend in the same direction. **Communication guidance:** Do not state that the effect is absent in THA. Rather, state that the effect was statistically significant in TKA but underpowered to detect in THA, where point estimates trended similarly. Avoid claiming a differential effect by surgery type without a formal interaction test.

VI. Sensitivity Analyses – Active Pharmacotherapy Only

Table 10. Sensitivity Analysis C – Active Pharmacotherapy Arms Only (Excluding Diet-Controlled; Firth-Penalized, Reference: Metformin)

Comparison	Outcome	aOR	95% CI
Insulin vs Metformin	Composite 90d	2.16	1.30–4.12
Insulin vs Metformin	Postop opioid	1.49	0.80–2.74
OtherDM vs Metformin	Composite 90d	3.72	1.74–9.11
OtherDM vs Metformin	Postop opioid	3.35	1.37–11.12

Restricted to patients receiving active diabetes pharmacotherapy (Metformin, Insulin, or Other DM drugs). Diet-Controlled arm excluded entirely. Same covariate adjustment.

Interpretation: Results are identical to the primary pairwise comparisons because the Firth pairwise design already compares each arm directly to Metformin. The value of this sensitivity analysis is conceptual: it confirms that findings hold when the comparison is restricted exclusively to patients on active drug therapy, eliminating any concern that the Diet-Controlled arm is driving results. **Communication guidance:** Present as evidence that the Insulin and OtherDM findings are not artifacts of including the large Diet-Controlled group. The drug-vs-drug comparison independently confirms metformin's favorable association.

VII. Sensitivity Analyses – Individual Composite Components

Table 11. Sensitivity Analysis D – Individual Components of the 90-Day Composite (Firth-Penalized, Reference: Metformin)

Component	Comparison	aOR	95% CI	Note
Infection (90d)	—	—	—	Too few events (n=1); analysis suppressed
Wound complication (90d)	—	—	—	Zero events; analysis suppressed
Reoperation (90d)	—	—	—	Zero events; analysis suppressed
VTE (90d)	DietControlled vs Metformin	4.47	2.38–8.23	Significant; other arms suppressed
Readmission (90d)	Insulin vs Metformin	1.85	1.02–3.75	Borderline significant
Readmission (90d)	OtherDM vs Metformin	4.36	2.15–11.97	Significant
Readmission (90d)	DietControlled vs Metformin	1.57	0.96–2.97	Non-significant trend
ED visit (90d)	Insulin vs Metformin	1.94	1.00–4.37	Borderline (CI includes 1.00)
ED visit (90d)	OtherDM vs Metformin	1.06	0.21–3.06	Non-significant
ED visit (90d)	DietControlled vs Metformin	1.27	0.63–2.88	Non-significant

Components analyzed individually with the same covariate adjustment as the primary analysis. Components with < 10 total events were not analyzed. Comparisons where individual arm event counts fell below 20 were suppressed for the non-reportable arms but computed where feasible.

Interpretation: Readmission is the primary driver of the composite outcome signal. OtherDM shows the strongest readmission association (aOR 4.36), and Insulin shows a borderline significant effect (aOR 1.85, lower CI 1.02). The VTE finding for DietControlled (aOR 4.47) is unexpected and should be interpreted cautiously—it may reflect the small number of VTE events overall, and the suppression of VTE data in other arms limits comparative inference. Infection, wound complications, and reoperation had too few events for individual analysis, justifying the composite approach. **Communication guidance:** State that readmission was the dominant component driving the composite signal. The VTE finding is exploratory and should be flagged as requiring validation in larger cohorts. Do not over-interpret borderline findings (ED visits for Insulin, aOR 1.94 with CI touching 1.00)—describe as ‘approaching significance’ or ‘trending’ rather than ‘significant.’

VIII. Summary Results Table

Table 12. Summary of Primary Outcomes with Unadjusted Rates and Adjusted Odds Ratios

Outcome	Metformin	Insulin	OtherDM	DietControlled
	Rate	Rate | aOR (95% CI) p	Rate | aOR (95% CI) p	Rate | aOR (95% CI) p
Composite 90d	23/121 (19%)	64/184 (35%) | 2.16 (1.30–4.12) p=0.002	21/46 (46%) | 3.72 (1.74–9.11) p=0.002	177/607 (29%) | 1.54 (0.96–2.77) p=0.076
Postop opioid	91/121 (75%)	151/184 (82%) | 1.49 (0.80–2.74) p=0.244	42/46 (91%) | 3.35 (1.37–11.12) p=0.004	361/607 (59%) | 0.31 (0.17–0.51) p=0.002

Metformin is the reference arm. aOR from Firth-penalized logistic regression with bootstrap 95% CI. Bolded p-values indicate statistical significance ($p < 0.05$).

Interpretation: This summary table encapsulates the core findings. Four of six pairwise comparisons reached statistical significance. The strongest signals are OtherDM's elevated complication risk (aOR 3.72) and DietControlled's reduced opioid use (aOR 0.31). Insulin shows a moderate but significant complication increase (aOR 2.16). **Communication guidance:** This is the key table for manuscript Results sections and conference presentations. Lead with the significant findings but always present the full table to avoid selective reporting. Frame the DietControlled opioid finding as expected (milder disease = less opioid burden) rather than a metformin-specific effect.

IX. Methodological Detail

Table 13. Drug Classification Counts in the Full Drug Exposure Table

Drug Class	Total Dispensations	Description
Other (non-DM, non-opioid)	2,719,793	All non-classified medications
Opioid	232,230	Opioid analgesics (used for outcome ascertainment)
Insulin	63,361	All insulin formulations (glargine, lispro, aspart, etc.)
Metformin	46,239	Metformin and brand-name equivalents
Other DM drugs	32,462	SU, SGLT2i, DPP-4i, TZD, meglitinides, AGI
GLP-1 receptor agonists	7,192	Semaglutide, liraglutide, dulaglutide, tirzepatide, etc.

Classification applied to all drug_exposure records in the cohort using string matching on drug concept names. Dispensation counts represent total records, not unique patients. A single patient may have multiple dispensation records across classes.

Interpretation: The drug classification pipeline captured 7,192 GLP-1 dispensations and 46,239 metformin dispensations across the full cohort. The large number of insulin records (63,361) reflects the chronic nature of insulin therapy with frequent refills. The 232,230 opioid records were used exclusively for outcome ascertainment (post-operative opioid use), not for exposure classification. **Communication guidance:** Present these counts as validation that the drug classification algorithm captured relevant medication classes. Note that dispensation counts reflect records, not unique patients, and that arm assignment was based on unique patient-level flags within the pre-operative window.

Table 14. Key Covariate Availability in the Analytic Cohort

Covariate	Available	Total	% Available	Imputation Method
HbA1c (%)	458	976	46.9%	Median imputation
BMI (kg/m²)	532	976	54.5%	Median imputation
Diabetes duration	~976	976	~100%	Derived from first T2DM diagnosis date; 0 if missing

HbA1c and BMI represent the most recent measurement within 365 days prior to surgery. Missing values were imputed with the cohort median for adjusted analyses. Diabetes duration calculated as years between first T2DM condition record and index surgery date.

Interpretation: Substantial missingness in HbA1c (53.1%) and BMI (45.5%) is a limitation. Median imputation is a simple approach that may attenuate associations if missingness is differential across arms (e.g., sicker patients may have more frequent lab monitoring). **Communication guidance:** Acknowledge HbA1c/BMI missingness as a study limitation. Note that this is a known characteristic of EHR-derived data in All of Us and does not necessarily indicate selection bias, but may reduce the precision of adjusted estimates. Future work could employ multiple imputation to assess sensitivity to this missingness.

End of Report. All tables represent the complete analytic output from the study pipeline. Cells with counts < 20 were suppressed per All of Us data use policy. Generated April 2026.